Krishil Patel

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Computer Science student focused on AI/ML, Generative AI, and agentic LLM systems, with experience building BERT/transformer NLP pipelines and LLM applications using LangChain, LangGraph, OpenAI SDK, and CrewAI. Strong Java/Spring Boot and AWS foundation for scalable ML services.

Education

Rutgers University - B.S. in Computer Science & Applied Physics — New Brunswick, NJ

Sep 2024 -- Present (Graduation: May 2027) — *Dean's List - GPA: 3.84 / 4.0*

- **CS Coursework:** Deep Learning, AI, ML, Data Structures, Algorithms, Computer Architecture, Databases, Numerical Analysis
- **Physics Coursework:** Modern Optics, Modern Physics, Advanced General Physics I & II, Analytical Physics IIb, Modern Instrumentation

Skills

- **AI / ML:** Python, Supervised & Unsupervised Learning, Classification, Feature Engineering, scikit-learn, BERT, Transformers, Hugging Face, NumPy
- **Generative AI:** LangChain, LangGraph, RAG, Embeddings, BM25, Hybrid Retrieval, Reranking, Chroma, Human-in-the-Loop
- **Data Analytics & Databases:** SQL, Pandas, Matplotlib, Seaborn, PostgreSQL, Amazon Aurora, SQLite, Flyway
- **Application Development:** Java, JavaScript, TypeScript, FastAPI, Pydantic, Spring Boot, REST APIs, Spring Data JPA, React, Vite, OAuth2, RBAC
- **Cloud:** AWS, Amazon Bedrock, Lambda, API Gateway, ECS Fargate, S3, Cognito, KMS, AWS CDK

Experience

Research Assistant - NJIT — Newark, NJ

Oct 2023 -- Jun 2024

- Processed and analyzed millions of Twitter/X and Reddit posts using Python to investigate social-media bias and geopolitical discourse, applying data cleaning, normalization, and feature engineering to large-scale unstructured datasets.
- Developed a BERT-based NLP pipeline for sentiment analysis and topic modeling, using tokenization, embeddings, and model fine-tuning to examine narrative framing, bias patterns, and shifts in public discourse.
- Conducted temporal, regional, and sentiment analyses with Pandas, Matplotlib, and Seaborn, translating large volumes of social-media data into interpretable research findings and visualizations.

Projects

RiskIQ: Agentic Risk Platform | *Python, FastAPI, LangGraph, DeepAgents, Cognee, Amazon Bedrock, React* — 2026

- Designed and built an agentic risk-operations platform with specialized AI workflows for fraud investigation, anomaly analysis, forecasting, model validation, workflow coordination, communications, and reporting.
- Orchestrated stateful agent workflows with FastAPI, LangGraph, DeepAgents, and Amazon Bedrock, incorporating human-in-the-loop review and approval controls for traceable AI-assisted decision making.
- Integrated Cognee for persistent agent memory and developed a React/TypeScript operations console for investigations, workflow status, execution traces, approval states, audit history, and explainable outputs.
- Defined structured outputs for root-cause analysis, fraud-risk reasoning, anomaly explanations, forecast-variance analysis, evidence synthesis, and audit-ready decision records for human review.

Role-Based Financial RAG Assistant | *Python, LangChain, Chroma, BM25, Hugging Face, RBAC, Gemini/OpenAI* — 2026

- Built a role-aware RAG assistant for banking refund and chargeback operations, enabling grounded natural-language Q&A across structured and unstructured financial content with source-cited responses.
- Combined Chroma vector retrieval, BM25 keyword search, and cross-encoder reranking to provide hybrid retrieval across heterogeneous banking knowledge sources.
- Enforced document-level authorization across 6 operational roles using RBAC and metadata governance, preventing restricted content from reaching LLM generation while preserving source-level traceability.
- Developed ingestion and validation workflows across 11 structured and unstructured file formats, with metadata-driven processing and repeatable dense and sparse indexing.

Adaptive AI Tutor | *Python, scikit-learn, LangChain, Chroma, SQLite, Gemini/OpenAI* — 2026

- Built an adaptive tutoring system combining supervised machine learning, RAG, and LLM generation to personalize instruction based on student performance and predicted concept mastery.
- Trained a logistic-regression mastery model using academic performance, engagement, practice history, and topic difficulty to estimate learning readiness and select explanation, worked-example, or quiz-based teaching strategies.
- Integrated LangChain and Chroma to ground tutor responses in retrieved course material, with SQLite persistence for student profiles, mastery predictions, and tutoring interactions across sessions.

Virtual Stock / Asset Management System | *Java, Spring Boot 3, React, AWS, PostgreSQL, Google Gemini* — 2025 -- 2026

- Architected and built a full-stack investment management platform using Java, Spring Boot, React, and PostgreSQL, exposing ~70 REST endpoints across portfolios, holdings, transactions, market data, analytics, authentication, and administration.
- Engineered scheduled market-data pipelines integrating 3 external providers across 4 asset types, with Amazon Aurora persistence, caching, time-series storage, and React/TypeScript analytics dashboards.
- Integrated Gemini with structured portfolio context to generate portfolio-specific financial analysis, and deployed the platform on AWS using containerized services, managed databases, authentication, encryption, and infrastructure as code with AWS CDK.

Certifications

- **Completed:** Oracle Java Foundations (1Z0-811)
- **In Progress:** AWS Certified Developer – Associate, Oracle Java SE 17 Developer (1Z0-829), Oracle Database SQL Certified Associate (1Z0-071)